

Advanced Coordinate Geometry

ANSWER KEY



Distance, midpoint, and slope combined

- 1 Triangle ABC has vertices $A(-2, 1)$, $B(4, 1)$, and $C(1, 7)$. Find the length of the median from C to the midpoint of AB .

Midpoint of AB : $\left(\frac{-2+4}{2}, \frac{1+1}{2}\right) = (1, 1)$. Length of median from $C(1, 7)$ to $(1, 1)$:
 $\sqrt{(1-1)^2 + (7-1)^2} = \sqrt{0+36} = 6$.

Equations of circles from given conditions

- 2 Find the equation of a circle with a diameter having endpoints $(2, 3)$ and $(8, 11)$.

Center (midpoint): $\left(\frac{2+8}{2}, \frac{3+11}{2}\right) = (5, 7)$. Radius (half the diameter length): diameter
 $= \sqrt{(8-2)^2 + (11-3)^2} = \sqrt{36+64} = \sqrt{100} = 10$, so radius $= 5$. Equation:
 $(x-5)^2 + (y-7)^2 = 25$.

Distance from a point to a line

- 3 Find the distance from the point $(3, 4)$ to the line $3x - 4y + 5 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
 $d = \frac{|3(3) - 4(4) + 5|}{\sqrt{9+16}} = \frac{|9-16+5|}{5} = \frac{|-2|}{5} = \frac{2}{5}$.
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Locus problems

- 4 Find the equation of the locus of points equidistant from $(2, 0)$ and $(-2, 0)$.

The locus of points equidistant from two fixed points is the perpendicular bisector of the segment joining them. Since both points lie on the x -axis symmetric about the origin, the perpendicular bisector is the vertical line $x = 0$ (the y -axis).

Rotations and general transformations

- 5 A point $(4, 0)$ is rotated 90° counterclockwise about the origin. Find the coordinates of the image point, using the rotation rule $(x, y) \rightarrow (-y, x)$.

Applying the rule: $(4, 0) \rightarrow (-0, 4) = (0, 4)$.

Dilations and scale factor

- 6 A triangle with vertices $(1, 2)$, $(3, 2)$, $(1, 6)$ is dilated by a scale factor of 3 centered at the origin. Find the coordinates of the image vertices.
Multiply each coordinate by 3: $(3, 6)$, $(9, 6)$, $(3, 18)$.
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Finding the equation of a parabola from three points

- 7 A parabola $y = ax^2 + bx + c$ passes through $(0, 3)$, $(1, 6)$, and $(-1, 2)$. Find a , b , and c .
From $(0, 3)$: $c = 3$. From $(1, 6)$: $a + b + c = 6$, so $a + b = 3$. From $(-1, 2)$: $a - b + c = 2$, so $a - b = -1$. Adding $a + b = 3$ and $a - b = -1$: $2a = 2$, so $a = 1$, then $b = 2$. Equation:
 $y = x^2 + 2x + 3$.
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A multi-part triangle-in-the-plane problem

- 8 This question has two parts. Triangle PQR has $P(0, 0)$, $Q(6, 0)$, and $R(3, 6)$. (a) Find the area of the triangle using the coordinate formula $\text{Area} = \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)|$. (b) Verify the area using the base-height formula instead.
(a) $\text{Area} = \frac{1}{2}|0(0 - 6) + 6(6 - 0) + 3(0 - 0)| = \frac{1}{2}|0 + 36 + 0| = 18$. (b) Using base $PQ = 6$ (along the x -axis) and height equal to R 's y -coordinate, 6: $\text{Area} = \frac{1}{2}(6)(6) = 18 \checkmark$, confirming both methods agree.
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Finding the equation of a line given a point and a perpendicular condition

- 9 Find the equation of the perpendicular bisector of the segment connecting $(2, 3)$ and $(8, 7)$.
Midpoint: $(\frac{2+8}{2}, \frac{3+7}{2}) = (5, 5)$. Slope of the segment: $\frac{7-3}{8-2} = \frac{4}{6} = \frac{2}{3}$. Perpendicular slope: $-\frac{3}{2}$.
Equation: $y - 5 = -\frac{3}{2}(x - 5)$, so $y = -\frac{3}{2}x + \frac{15}{2} + 5 = -\frac{3}{2}x + \frac{25}{2}$.
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Translating and reflecting equations of curves

- 10 The graph of $y = x^2$ is reflected over the x -axis, then shifted 3 units up. Write the equation of the resulting graph.
Reflecting over the x -axis: $y = -x^2$. Shifting up 3: $y = -x^2 + 3$.
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Finding vertices of a polygon given transformations

- 11 A square has vertices $(0, 0)$, $(4, 0)$, $(4, 4)$, $(0, 4)$. Find the coordinates after translating the square 3 units right and 2 units down.
Add $(3, -2)$ to each vertex: $(3, -2)$, $(7, -2)$, $(7, 2)$, $(3, 2)$.
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Finding the intersection of a line and a parabola

- 12 Find the points of intersection of $y = x^2 - 2x - 3$ and $y = x - 3$.

Set equal: $x^2 - 2x - 3 = x - 3$, so $x^2 - 3x = 0$, giving $x(x - 3) = 0$, so $x = 0$ or $x = 3$. Corresponding y -values: for $x = 0$, $y = -3$; for $x = 3$, $y = 0$. Intersection points: $(0, -3)$ and $(3, 0)$.

A multi-part quadrilateral classification problem

- 13 This question has three parts. Quadrilateral $WXYZ$ has vertices $W(0, 0)$, $X(6, 0)$, $Y(8, 4)$, $Z(2, 4)$. (a) Find the length of each side. (b) Find the slopes of WX and ZY , and of WZ and XY . (c) Based on parts (a) and (b), classify the quadrilateral as precisely as possible.

(a) $WX = \sqrt{(6-0)^2 + 0^2} = 6$. $XY = \sqrt{(8-6)^2 + 4^2} = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5}$.

$YZ = \sqrt{(2-8)^2 + 0^2} = 6$. $ZW = \sqrt{(0-2)^2 + (-4)^2} = \sqrt{4+16} = \sqrt{20} = 2\sqrt{5}$. (b) Slope of WX :

$\frac{0-0}{6-0} = 0$. Slope of ZY : $\frac{4-4}{8-2} = 0$ (parallel, since equal slopes). Slope of WZ : $\frac{4-0}{2-0} = 2$. Slope of XY :

$\frac{4-0}{8-6} = 2$ (also parallel, equal slopes). (c) Since both pairs of opposite sides are parallel ($WX \parallel ZY$ and $WZ \parallel XY$) and opposite sides are equal in length ($WX = YZ = 6$, $XY = ZW = 2\sqrt{5}$), the quadrilateral is a parallelogram.

Finding the equation of a tangent line to a circle

- 14 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.

The radius to $(3, 4)$ has slope $\frac{4-0}{3-0} = \frac{4}{3}$. The tangent line is perpendicular, so its slope is $-\frac{3}{4}$. Equation: $y - 4 = -\frac{3}{4}(x - 3)$, so $y = -\frac{3}{4}x + \frac{9}{4} + 4 = -\frac{3}{4}x + \frac{25}{4}$.

Finding the vertex and axis of symmetry of a parabola from three points

- 15 A parabola passes through $(-1, 0)$, $(3, 0)$, and $(1, -8)$. Find the axis of symmetry and the vertex.

Since the parabola has zeros at $x = -1$ and $x = 3$, the axis of symmetry lies midway between them:

$x = \frac{-1+3}{2} = 1$. Since $(1, -8)$ lies exactly on the axis of symmetry $x = 1$, this point IS the vertex: $(1, -8)$.